

$$(T_1^2 + T_2^2 + T_3^2)|v\rangle = 0$$

Assume that each T_i is self-adjoint, i.e.

$$T_i^\dagger = T_i \quad \text{for } i = 1, 2, 3.$$

Taking the inner product with $\langle v|$, we get

$$\langle v|(T_1^2 + T_2^2 + T_3^2)|v\rangle = 0.$$

By linearity,

$$\langle v|T_1^2|v\rangle + \langle v|T_2^2|v\rangle + \langle v|T_3^2|v\rangle = 0.$$

Since $T_i^\dagger = T_i$, we have

$$\langle v|T_i^2|v\rangle = \langle v|T_i^\dagger T_i|v\rangle = \langle T_i v|T_i v\rangle = \|T_i v\|^2.$$

Therefore,

$$\|T_1 v\|^2 + \|T_2 v\|^2 + \|T_3 v\|^2 = 0.$$

Each term is nonnegative, so each term must be zero:

$$\|T_1 v\|^2 = 0, \quad \|T_2 v\|^2 = 0, \quad \|T_3 v\|^2 = 0.$$

Hence,

$$T_1 v = 0, \quad T_2 v = 0, \quad T_3 v = 0.$$

Thus,

$$(T_1^2 + T_2^2 + T_3^2)|v\rangle = 0 \implies T_1 v = T_2 v = T_3 v = 0,$$

provided T_1, T_2, T_3 are self-adjoint.